

BEYOND REFERENCE MATCHING: SPECIFICATION-BASED CORRECTNESS EVALUATION FOR DSP IMPLEMENTATIONS

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ABSTRACT

A reference implementation is commonly used as the correctness target for DSP implementations. That comparison is exact for a unique map, but a magnitude mask defines a feasible set, not one coefficient vector. We score a candidate by specification-set membership ($S_t = 1$ if it meets a frozen contract) and compare coefficient and spec-band response agreement to one canonical design. On 20 non-unique mask instances, reference matching rejects 374/416 constructed-valid implementations ($\text{FRR}_{\text{ref}} = 0.899$; 0.927 on tight masks). On eight singleton identities, $\text{FRR}_{\text{ref}} = 0$. Against the same constructed labels, the coefficient oracle has $\text{FRR} = 0.899$ and $\text{FAR} = 0$, and spec-band similarity has $\text{FRR} = 0.067$ and $\text{FAR} = 0$; membership $\text{FRR} = \text{FAR} = 0$ restates the admission rule. The same $S_t = 1$, $R_{t,r} = 0$ event appears among existing generated implementations. On these non-unique specifications, reference agreement is a realization diagnostic. Our code is available at: <https://github.com/XianghuiMeng-1020/dsp-semantic-correctness>.

Index Terms— Filter design, specification testing, correctness evaluation, FIR, IIR

1. INTRODUCTION

Evaluating a DSP implementation against a trusted reference is natural when the specification is a unique map: one integer delay, one declared Fourier convention, one alias-fold period. In those cases, coefficient or output agreement with the reference *is* correctness.

Classical filter design is different. A magnitude mask with a free transition defines a *feasible set* of realizations, not a unique impulse response [1, 2, 3, 4]. Windowed designs [5, 6], frequency sampling [7], equiripple Parks–McClellan approximation [8, 9, 10], and IIR Butterworth or elliptic sections [11, 12, 13, 14] can all meet the same passband, stopband, and stability bounds while remaining far apart as coefficient vectors. A canonical library call—Hamming `firwin`, or a Butterworth section [15]—is one occupant of that set.

Therefore a valid reference cannot, by itself, define correctness over the whole set. Matching that reference asks whether a candidate is the *same realization*. It does not ask whether the candidate meets the contract that justified the reference.

This paper treats that distinction as an evaluation problem for generated DSP implementations [16, 17], not as a model leaderboard and not as a new design algorithm. The object of study is the scoring rule, not model quality.

Formally, reference agreement $R_{t,r}(h) = 1$ asks whether h matches a particular realization h_r . Specification membership $S_t(h) = 1$ asks whether h meets the frozen contract. When the feasible set $\mathcal{V}_t = \{h : S_t(h) = 1\}$ contains more than one realization, these questions diverge: $R_{t,r}$ is a realization diagnostic, and S_t is the correctness criterion used below. When the specification is effectively a singleton under the declared scoring tolerance, they should agree.

Related work falls in three clusters. Filter-approximation texts treat the mask as a feasible-set problem [1, 10, 18]. DSP algorithm tests may use a golden reference [19]. A standard generated-code protocol uses unique-output unit tests [17]; that protocol does not score a non-unique DSP mask. We do not survey model families.

We make three contributions. (i) We distinguish realization agreement from specification-set membership as scoring rules for DSP implementations. (ii) We construct a singleton suite and a non-unique magnitude-mask suite with frozen valid and invalid labels, independent of reference distance. (iii) We measure the evaluation consequence: reference false rejection is high on the constructed non-unique suite, zero on the singleton suite, and the same $S_t = 1$, $R_{t,r} = 0$ event appears among existing generated implementations.

2. PROBLEM FORMULATION

A design task t asks for an implementation h (FIR taps, or an IIR pair (b, a)). A frozen specification predicate returns $S_t(h) \in \{0, 1\}$. The feasible set is

$$\mathcal{V}_t = \{h : S_t(h) = 1\}. \quad (1)$$

A canonical reference h_r is one occupant, $h_r \in \mathcal{V}_t$, and nothing stronger: it is a convenient valid realization, not the definition of t .

The task is treated as a *singleton* when the declared map is unique up to the scoring tolerance (unique lag, unique DFT convention, unique alias map, unique impulse). It is *non-unique* when more than one realization meets S_t . The non-unique family studied here is a classical magnitude mask: constrained pass and stop bands on $|H(f)|$, a free transition, and, for IIR, a pole-radius stability bound. Linear phase is not required.

3. SPECIFICATION-BASED EVALUATION

Write $\tilde{h}$ for the coefficient vector of h , or the concatenation (b, a) , truncated to the shorter length when orders differ. Oracle A (reference coefficients) is

$$R_{t,r}(h) = \mathbf{1} \left[\frac{\|\tilde{h} - \tilde{h}_r\|_2}{\|\tilde{h}_r\|_2} \leq \tau_R \right], \quad (2)$$

with frozen $\tau_R = 0.05$. $R_{t,r}(h) = 1$ asks whether h is the same realization as h_r .

Oracle B (response similarity) accepts h when the $|H|$ RMSE on constrained bands versus h_r does not exceed the maximum same-order library-pair RMSE on that instance, plus 10^{-8} . Oracle B is a DSP-native comparison to *one* element of $\mathcal{V}_t$. It is not a gold standard for correctness.

Oracle C implements the same predicate S_t used to admit constructed labels; it is the membership checker, not an external labeler. On each constrained band $[f_0, f_1]$ with bounds $[\ell, u]$, a 4096-point `freqz` residual records the worst normalized excursion of $|H|$ outside $[\ell, u]$. FIR contracts pass at residual $\leq 10^{-6}$; IIR contracts also require every pole radius < 0.999 and pass at residual $\leq 10^{-3}$. Neither floor was fit to generated residuals. The checker does not load h_r , does not compute coefficient distance, and does not call Oracle A or B. A tone-envelope consistency check T reads the same mask and is not used as gold for FAR/FRR.

Fig. 1 states the geometry: the τ_R -ball around h_r is a realization neighborhood; $\mathcal{V}_t$ is the contract.

4. EXPERIMENTAL PROTOCOL

Two suites are scored (Table 1). Suite S comprises eight singleton identities already used as unique-map contracts (integer delay, circular and linear convolution conventions, lag-0 energy, alias fold, frequency rescale, Nyquist frequency, and $\delta[n - D]$). Suite N comprises 20 specification instances from a common magnitude-mask family: FIR low-pass, high-pass, band-pass, and band-stop at $f_s = 8$ and 16 kHz, each at a loose and a tight mask (16 FIR), plus IIR low-pass and high-pass at 8 kHz (loose/tight; 4 IIR). Loose 8 kHz prototypes reuse the previously frozen pass/stop numbers. Tight masks

Table 1. Study suites. Arm G is a witness on four original masks and is not added to the constructed counts.

Suite	Inst.	Spec. type	Uniq.	Valid	Invalid	Witness
S	8	unique map	yes	12	16	—
N	20	mag. mask	no	416	144	Arm G, 4

are derived mechanically (pass ripple and stop ceiling $\times 1/5$; stop edge moved halfway toward the facing pass). Sixteen-kilohertz edges are the 8 kHz edges $\times 2$. Canonical h_r is the shortest-odd Hamming firwin (FIR) or the lowest-order butter (IIR) that meets S_t .

Suite N valids are constructed from three sources: classical library designers (firwin, firwin2, remez, firls, Kaiser firwin; butter, cheby1, cheby2, ellip); first-principles FIR windowed-sinc and frequency sampling (numpy only; SciPy design APIs forbidden) [5, 7]; and random-valid search over odd length, window or Remez weights, and cutoffs in the free transition (IIR: prototype and order). Admission is $S_t(h) = 1$ only. Distance to h_r is measured after admission. Occupants that met S_t and lay near h_r were retained. The random-valid search does not maximize coefficient distance, response RMSE, or disagreement with h_r . Near-duplicates are removed ($d_{\text{coeff}} \leq 0.01$ and full-grid $|H|$ RMSE $\leq 10^{-3}$); deduplication does not maximize distance. The kept set has 416 valids (92 library, 24 first-principles, 300 random-valid).

Invalids are 144 mechanism-based contract violations (band swap, cutoff into a constrained band, insufficient order, unstable pole, Nyquist coded as 1 Hz, pass-gain collapse, wrong f_s , and type mismatch), not a sample of naturally occurring errors. A mutant is admitted only if $S_t = 0$. Labels were frozen before any oracle comparison.

Arm G is a generated-code witness on the original four loose 8 kHz masks only (48 planned draws; no new generations). It is not pooled into the constructed n . Constructed occupants score the oracles; Arm G is only an application witness.

Primary rates use constructed labels. FRR_{ref} is the fraction of valids with $R_{t,r} = 0$. $\text{FRR}_{\text{ref}}^{\text{any}^3}$ rejects a valid only if it is discordant with every available library occupant among firwin/remez/firls (FIR) or the IIR library set. Oracles A–C are scored as FRR/FAR against the same labels. Suite S uses the same definitions on identity outputs.

5. RESULTS

RQ1 — Reference false rejection. On Suite S, $\text{FRR}_{\text{ref}} = 0/12$. On Suite N, reference matching rejects 374/416 constructed-valid implementations ($\text{FRR}_{\text{ref}} = 0.899$; Table 2). The rate is 302/340 = 0.888 on FIR, 72/76 = 0.947 on IIR, 183/210 = 0.871 on loose masks, and 191/206 = 0.927 on tight masks. Every one of the 20 instances has at least one valid with $S_t = 1$ and $R_{t,r} = 0$. Best-of-three refer-

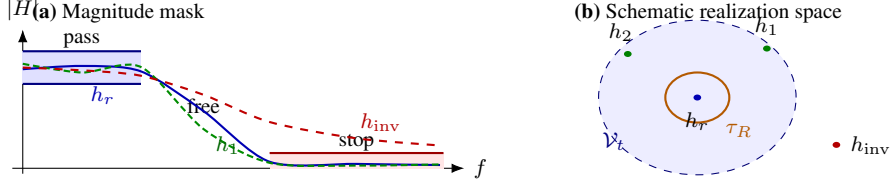


Fig. 1. Same contract, two spaces. (a) Frequency-domain magnitude mask. (b) Schematic realization/coefficient space: the τ_R -ball is proximity to one valid h_r ; the $\mathcal{V}_t$ region is schematic and is not a Euclidean ball in the same metric as τ_R . Occupants h_1, h_2 meet the mask but lie outside the reference ball; h_{inv} violates the contract.

Table 2. Reference rejection at $\tau_R = 0.05$ on constructed-valid occupants (n). Task-level disagreement: instances with at least one constructed valid that fails Oracle A while meeting the contract.

Suite	n	FRR_{ref}	$\text{FRR}_{\text{ref}}^{\text{any}^3}$	Task disag.
S	12	0/12 = 0	—	—
N all	416	374/416 = 0.899	346/416 = 0.832	20/20
N FIR	340	302/340 = 0.888	274/340 = 0.806	—
N IIR	76	72/76 = 0.947	72/76 = 0.947	—
N loose	210	183/210 = 0.871	177/210 = 0.843	—
N tight	206	191/206 = 0.927	169/206 = 0.820	—

ences still reject $346/416 = 0.832$. These fractions describe disagreement inside constructed feasible sets. They are not rates over DSP implementations at large, and they are not rates over generated code.

RQ2 — Diversity of $\mathcal{V}_t$. Library, first-principles, and random-valid sources all contribute occupants with $S_t = 1$. On every Suite N instance the median min-length coefficient distance to h_r is at least 1.0, while median spec-band $|H|$ RMSE stays on the order of 10^{-3} (FIR) to 10^{-2} (IIR); full-grid RMSE is larger because the transition is unconstrained. First-principles windowed-sinc and frequency-sampling designs enter the same sets without calling a library designer, so multiplicity is not an API artifact. One concrete pattern: a same-order first-principles occupant can remain inside the mask with spec-band RMSE $\sim 10^{-3}$ while $d_{\text{coeff}} > \tau_R$ versus Hamming *firwin*. On the 67 occupants whose order equals that of h_r , $\text{FRR}_{\text{ref}} = 25/67 = 0.373$, versus 0.899 on the unrestricted 416 (Table 4). Order variation contributes materially to coefficient disagreement. A nonzero rate remains on this restricted subset: realization equality is still not equivalent to contract membership. All constructed FIR occupants in this set are Type I, so a linear-phase restriction is not a separate ablation here. Varying $\tau_R \in \{0.01, 0.05, 0.10\}$ changes FRR_{ref} only from 0.933 to 0.873 (Table 4).

RQ3 — Oracles versus constructed labels. Table 3 compares A–C on the frozen contract-defined universe. Against those labels, Oracle A has $\text{FRR} = 0.899$ and $\text{FAR} = 0$: it never accepts a mutant, and it rejects most valids. Oracle B has $\text{FRR} = 0.067$ and $\text{FAR} = 0$: most constructed valids stay close to h_r on the constrained bands, so B rarely

Table 3. FRR/FAR on the contract-defined constructed universe (N: 416 valid, 144 invalid; S: 12 valid, 16 invalid). Oracle C implements S_t , so its zeros check label consistency, not independent validation. The informative contrast is A and B on the same labels. Oracle B is undefined on Suite S.

Suite	Oracle (rule)	FRR	FAR
N	A (coeff. vs. h_r , $\tau_R = 0.05$)	0.899	0
N	B (band $ H $ vs. h_r)	0.067	0
N	C (specification membership)	0	0
S	A (output vs. canonical)	0	0
S	C (specification membership)	0	0

false-rejects even when the taps differ. That agreement does not make B a membership test: it still scores proximity to one realization. Oracle C has $\text{FRR} = \text{FAR} = 0$ because it implements the same S_t used to write the labels; the row is implementation consistency, not external validation of C. The informative comparison is how A and B reject subsets of the same contract-valid set. A asks realization identity. B asks response proximity to one realization. C asks contract satisfaction.

RQ4 — Singleton contrast. On Suite S, Oracles A and C both have $\text{FRR} = \text{FAR} = 0$. The argument is conditional. When the specification is a unique map, reference matching and specification membership coincide, and reference matching is the appropriate correctness test. The Suite N false-rejection rates are not a generic indictment of reference comparison.

RQ5 — Generated-code witness. Among the existing eligible generated implementations on the original four loose 8 kHz masks, nine satisfied the frozen specification while failing reference agreement. Such witnesses occurred on all four of those specification instances. This is an existence witness that the scoring mismatch is not confined to constructed filters. It is not a prevalence estimate and not a model comparison.

6. DISCUSSION

Reference matching is not universally wrong. It is the right correctness test on Suite S, where the specification is effectively a singleton. On Suite N it measures proximity to one

Table 4. Sensitivity of FRR_{ref} on Suite N constructed valids. The τ_R rows use all 416 occupants. The order-lock row uses only the 67 occupants whose order equals that of h_r . Thresholds were not retuned after seeing outcomes.

Knob	n	FRR_{ref}
$\tau_R = 0.01$	416	$388/416 = 0.933$
$\tau_R = 0.05$	416	$374/416 = 0.899$
$\tau_R = 0.10$	416	$363/416 = 0.873$
order = h_r	67	$25/67 = 0.373$

valid realization rather than membership in $\mathcal{V}_t$. Tightening these frozen masks did not eliminate the disagreement (tight $\text{FRR}_{\text{ref}} = 0.927$). On the canonical-order subset, FRR_{ref} is 0.373 (25/67), versus 0.899 on all 416: order contributes materially, but a same-order valid can still fail Oracle A while meeting the contract. Oracle C’s zeros are consistency of the membership checker with the constructed labels. The substantive comparison is that A and B reject different subsets of that same universe. Oracle B can mostly agree with C while still asking a different question: spec-band $|H|$ may stay close to one h_r when the taps do not.

The study is restricted to magnitude-mask filter-design specifications. It is not a claim about beams, adaptive filters, filter banks, or other DSP families. Requiring Type-I linear phase, or a declared maximum order, would be a *different* specification and a different feasible set; we do not treat those constraints as the default contract. Arm G is an existence witness on four original masks, not evidence of how often generated code is correct, and it is not pooled with the constructed n used to score A–C. Six Suite N instances had no same-order library pair for Oracle B; those thresholds used the all-library pairwise maximum.

7. CONCLUSION

When a DSP specification defines a feasible set rather than a single realization, the correctness test used here is membership in that set. Reference matching remains useful as a realization diagnostic and remains appropriate for singleton specifications.

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